Is Twitch Sustainable?: Streaming for Sustainability

Dissertation in partial fulfilment of the requirements for the degree of

MASTER OF SCIENCE WITH A MAJOR IN MACHINE LEARNING AND STATISTICS



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Quantifying the Carbon Footprint of Twitch Streaming: A Controlled Experiment and Real-World Analysis

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Abstract

This thesis investigates the environmental impact of the live-streaming platform Twitch.tv, specifically focusing on its carbon emissions. While the environmental impact of video streaming as a whole has been investigated, there is limited research specifically quantifying the carbon emissions associated with the Twitch platform.

This study addresses this gap by employing a two-pronged approach. First, controlled experiments were conducted on Amazon Web Services (AWS) infrastructure, which underpins much of Twitch's operations, to directly measure carbon emissions at various streaming quality levels. Second, data was collected from the Twitch API to understand real-world usage patterns among streamers, by correlating these two datasets to estimate the carbon footprint of typical Twitch streams and identify key factors, such as stream duration, quality, and viewership.

This research provides a foundational understanding of Twitch's environmental impact. It offers a framework for developing targeted strategies to enhance the platform's sustainability within the broader context of the streaming industry.

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1 Introduction

The digital age has ushered in an era of unprecedented convenience and connectivity, fundamentally transforming how we consume entertainment. Online streaming platforms like Twitch, YouTube, and Netflix have become integral to our daily lives, offering a vast array of content at our fingertips. However, this digital revolution has not come without its environmental costs. The substantial energy demands of streaming infrastructure, coupled with the growing user base and content consumption, have raised concerns about the carbon footprint of our digital entertainment choices.

In 2018, video streaming alone contributed 300 million tonnes of CO_2 emissions, a figure equivalent to the annual emissions of France[1]. This statistic underscores the urgent need to scrutinize the environmental implications of our streaming habits. Among the various streaming platforms, Twitch stands out as a dominant force in the realm of live streaming. With over 30 million daily active users and an average of 1.5 million concurrent viewers at any given time, Twitch provides a dynamic platform for a diverse range of content creators and viewers. From video game broadcasts and esports tournaments to live music performances and creative content, Twitch has cultivated a vibrant online community.

However, the growing popularity of Twitch and the increasing demand for high-quality live streams have amplified concerns regarding its environmental impact. The platform's energy consumption is not solely attributed to streaming itself. It extends to the vast network of data centres and servers that underpin its operations. Data centres, which house the hardware necessary for storing, processing, and delivering content, are notorious for their energy-intensive nature. In 2020, data centres were estimated to consume 1.8% of global electricity demand, and this figure is projected to rise further as the demand for digital services continues to surge.

The environmental impact of Twitch further encompasses factors beyond data centre energy consumption. The development of video games, a primary source of content on the platform, involves substantial energy use and resource consumption. Additionally, travel for live events, the manufacturing of streaming devices such as high-end gaming PCs and smartphones, and the energy required to power these devices during streaming sessions all contribute to the platform's broader environmental burden.

Understanding the environmental consequences of our digital choices, particularly within platforms like Twitch, necessitates a comprehensive examination of the interplay

between usability, design, and sustainability. The pursuit of seamless user experiences often translates to high-resolution streams, resource-intensive features, and continuous content delivery. While these elements are intended to enhance user engagement and satisfaction, it's important to consider their potential to inadvertently amplify the environmental costs associated with streaming.

To address these concerns and promote a more sustainable approach to digital entertainment, it is crucial to quantify the carbon footprint of Twitch streams and identify the factors that influence their environmental impact. This research aims to answer several key questions:

1. What is the average carbon footprint of a Twitch stream?

2. How do specific stream characteristics influence carbon emissions?

- **Stream Duration:** Is there a direct correlation between the length of a stream and its carbon emissions? Does a longer stream always equate to larger carbon emissions?
- **Video Quality:** How significant is the difference in carbon emissions between a high-quality stream and a lower-quality alternative?
- **Viewership:** Does the number of viewers watching a stream directly impact its carbon footprint? Is there a linear relationship between audience size and emissions?

By addressing these questions, this research seeks to shed light on the environmental impact of Twitch streaming and provide insights into potential strategies for mitigating its carbon footprint. This information will empower content creators, viewers, and platform developers to make more informed decisions about their digital entertainment choices and contribute to a more sustainable future for online streaming.

The research was conducted within a controlled environment to ensure accurate and reliable results. This controlled setting will allow for the isolation of specific variables and the precise measurement of their impact on carbon emissions. The relationship between these variables and the resulting carbon footprint can be established by manipulating factors such as stream duration, video quality, and viewership.

The findings of this research will have significant implications for understanding and potentially mitigating the environmental impact of live-streaming platforms. By quantifying the carbon footprint of Twitch streams on the server side and identifying the key factors contributing to their environmental impact, this study will provide valuable insights that can inform decision-making for both platform providers and users. Content creators can utilize this information to optimize their streaming practices and minimize their environmental impact. Viewers can make more informed decisions about the content they consume and the platforms they support. Platform developers can leverage

these findings to implement more sustainable infrastructure and features, ultimately leading to a more environmentally responsible approach to online streaming.

2 Literature Review

In this section, a review of relevant literature relating to the topic of this project is given as well as a discussion and evaluation of existing research within the context of this project's scope.

2.1 Energy emissions of ICT and data centres

The growing awareness of climate change has intensified scrutiny of the carbon impact of the global Information and Communications Technology (ICT) sector, a sector that has expanded rapidly in recent decades due to the increasing digitization of human activities. This has led to numerous studies attempting to quantify the ICT sector's global carbon emissions. However, measuring these emissions is inherently complex due to the difficulty of defining clear boundaries within the sector. The internet, a fundamental component of the ICT sector[1], is a vast and interconnected network, making it challenging to establish a universally accepted boundary for analysis.

Furthermore, the ICT sector comprises various sub-sectors, including manufacturing, network operations, data centres, and IT services. This diversity raises questions about the scope of carbon footprint assessments, such as whether to include embodied emissions from ICT equipment manufacturing. Additionally, the carbon intensity of electricity consumption[1]–[3], which varies across time and location, further complicates emission calculations.

These complexities have led to a range of estimates for the ICT sector's carbon footprint. Some studies[4], [5] suggest it accounts for 1.8-2.8% of the global footprint, while more assessments[6] place it closer to 1.3%, or 0.7 billion tonnes of CO_2 equivalent (CO_2eq).[1] It's important to note that these figures often include embodied emissions and are primarily estimates or predictions[1], [7].

Within the ICT sector, data centres contribute less than 20% of total emissions, with end-user devices being the primary source. The Carbon Trust[1], for example, estimates that data centres account for 141 MtCO_2eq of the sector's emissions, while end-user devices contribute 401 MtCO_2eq.

[1] Carbon dioxide equivalent (CO_2eq, CO_2e or CO_2-e) is a way to measure and compare the greenhouse gas emissions of different gases based on their global warming potential (GWP) over a 100 years, where carbon dioxide is used as the gas reference.

Despite this evidence, data centres have faced disproportionate negative attention from the public and media, with some sources claiming their emissions are considerably higher than reported in peer-reviewed journals. This discrepancy likely arises from a lack of public understanding regarding data centre energy usage and the media's tendency to oversimplify or misinterpret research findings.

Adding to the challenge of accurate assessment is the lack of transparency from many cloud service providers and large companies that own data centres regarding their energy consumption and carbon emissions[8]. This opacity not only contributes to uncertainty about the actual global emissions but also hinders their clients from accurately measuring their carbon footprints. However, there are positive developments, such as Microsoft's Sustainability Calculator, which provides GHG Protocol emissions reports for their enterprise cloud customers. Similar tools have also been introduced by Amazon and Google Cloud, signalling a move towards greater transparency[9], [10].

2.1.1 Data Center Growth

A notable trend in recent years is the decoupling of data centre energy usage from the exponential growth of internet traffic and workloads. Despite substantial increases in demand, data centre energy consumption has remained relatively stable[1]. This remarkable achievement can be attributed, in part, to Moore's Law, which observes the doubling of transistors in integrated circuits every two years[11]. This principle has driven continuous innovation and efficiency improvements in the semiconductor industry, leading to more powerful and energy-efficient hardware[12].

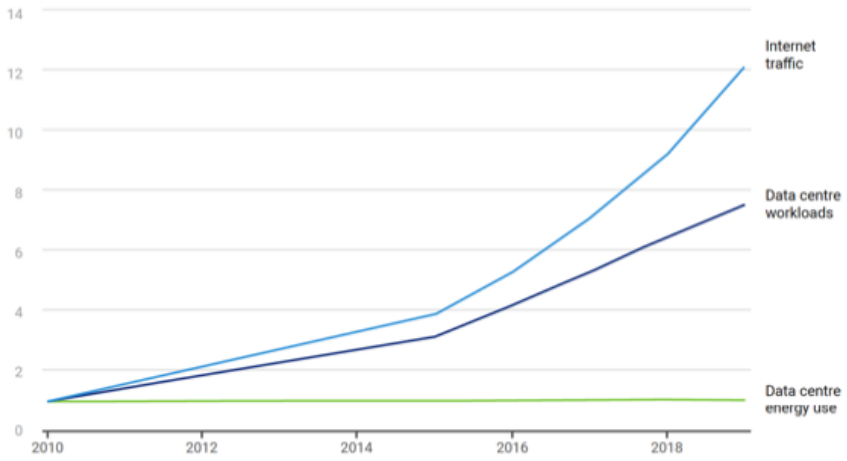


Figure 2.1: Global Internet traffic, data centre energy and workload trends for 2015-2019. Source: IEA 2020[1]

The ongoing advancements in technology, coupled with the implementation of energy-efficient practices within data centres, have allowed the industry to meet the growing demands of the digital age while mitigating its environmental impact. However,

continued efforts are necessary to further optimize energy usage and reduce the carbon footprint of data centres as the demand for digital services continues to rise[5].

2.2 Energy Emissions of Streaming

The proliferation of high-speed internet and the rise of streaming services like Netflix, YouTube, and Twitch have led to growth in video streaming. The increasing prevalence of video streaming has prompted inquiries into the environmental implications of this digital activity. However, it is essential to evaluate the claims of the media, which may not align with research findings regarding the actual carbon footprint of video streaming. Research by the International Energy Agency (IEA)[13] indicates that the carbon footprint of video streaming remains relatively modest compared to other activities. In 2019, the IEA estimated that one hour of video streaming produced an average of 36gCO2eq[13]. This figure is significantly lower than some media-reported estimates, which may be based on outdated or inaccurate data.

The actual energy consumption and associated carbon emissions of video streaming depend on several factors, including the user's device, network connection, and the resolution of the video. End-user devices, such as smartphones and computers, typically consume the most energy during streaming. The quality of the network connection also plays a role, with higher-quality connections requiring more energy[14].

Content Delivery Networks (CDNs) have become a critical component of the streaming ecosystem[15]. By caching and delivering content closer to users, CDNs reduce the distance data needs to travel, thereby improving energy efficiency. The number of simultaneous streams[16] also affects energy consumption, with estimates suggesting that each stream consumes an average of 0.8 kWh.

Regional variations in energy sources and carbon intensity further influence the carbon footprint of streaming. Countries with a higher reliance on renewable energy sources, such as Sweden, will have lower carbon emissions per stream compared to countries with a higher proportion of fossil fuels in their energy mix.

2.3 Research Gap and Objectives

While existing research provides valuable insights into the energy consumption of data centres, networks, and the broader ICT sector[1], a crucial gap remains in the literature: the lack of studies directly quantifying the carbon emissions associated with Twitch. This platform-specific data is essential for understanding Twitch's unique carbon footprint and developing targeted strategies to mitigate its environmental impact.

This research aims to address this gap by employing a two-pronged approach. Firstly, controlled experiments on Amazon Web Services (AWS) will directly measure the carbon emissions of Twitch streams under varying conditions, including different video qualities and stream durations[9]. By leveraging AWS's detailed energy usage reports and regional carbon intensity factors, precise estimates of the carbon emissions associated with each stream can be derived. Secondly, the Twitch API[17] is utilized to gather and analyze real-world streaming data, providing insights into user behaviour, content consumption patterns, and the distribution of stream characteristics across the platform.

By combining controlled experiments with real-world data analysis, this research will offer a comprehensive assessment of Twitch's infrastructural carbon footprint. This information will be instrumental in identifying key areas where modifications in user choices, platform features, or infrastructure could lead to emissions reductions. For instance, if high-quality streams are found to be disproportionately carbon-intensive, the platform could explore incentivizing lower-quality options or developing more efficient encoding technologies.

Ultimately, this research aims to contribute to a more sustainable future for live-streaming platforms by providing actionable insights and fostering a more informed dialogue about the environmental impact of our digital entertainment choices. By quantifying the carbon emissions associated with Twitch and understanding the factors that influence them, this study will empower stakeholders to make informed decisions and take meaningful steps towards reducing the platform's carbon footprint.

3 Background

This chapter will delve into this project's theoretical and conceptual underpinnings and explain all of the concepts mentioned.

3.1 Data centres

Data centres play a crucial role in modern life, facilitating a wide range of daily activities, from streaming videos to online transactions. While their significance might not always be apparent to the average individual, data centres are fundamental to the functioning of the internet and the digital services we rely on It has been reported that a person interacts with a data centre roughly 40 times a day[18]. Without data centres, everyday tasks such as streaming videos, having food delivered, paying for the subway, remote calls, and so many more activities would not be possible. According to a report from the Renewable Energy Policy Project on Energy Smart Data Centres quoted in Data Centres Fundamentals [19] "data centres are an essential component of the infrastructure supporting the Internet and the digital commerce and electronic communication sector".

In its simplest form, a data centre is a physical space, typically enclosed, designed to house computing resources in a controlled and secure environment to support various Information Technology (IT) operations for a range of users and industries. These users and industries include organisations such as businesses, governments, researchers and universities to name a few. Data centres provide these organisations with the necessary infrastructure to store and process data, facilitate telecommunications, build, run and deliver applications and services and more. The computing resources found in a data centre include the three SSS: Servers, Switches and Storage.

3.1.1 History of data centres

The evolution of data centres can first be attributed to the start of mainframe computers and their maintenance. Most early computers were immensely larger and required a large space. Due to the complexities involved with maintenance and operation, it became a practice to keep them in dedicated rooms [20]. These computers had many cables and required tight security and maintenance, elevated floors, cooling systems and power. In the 1950s and 1960s, many examples of early computer centres were coming into existence within a variety of sectors, for example, in the banking industry, medical sector and the military [20].

In the 1980s there was a boom in the microcomputer industry with computers being deployed everywhere. However, with the growing complexities of IT operations, there became a need for companies to take control of IT resources. The 1990s saw the emergence of client-server computing with new micro-servers ending up stored in old computer rooms. The inexpensive networking equipment and new network cabling standards allowed for servers to be placed in a specific room inside the company. The term data centre for these specially designed computer rooms also soon started gaining traction [21].

Then came the dot-com boom in the 1990s which accelerated the use of data centres. Companies began to realise the importance of online presence and the need for nonstop operations to be deployed, which meant these companies needed faster internet connectivity to support such activities. However, it was not viable for smaller companies to install essential equipment to maintain their online presence. This pushed companies to build large facilities, Internet Data Centres (IDCs). This led to new technological advances in scaling operations and solutions for system deployments, all of which now make up the common practices seen today in data centres [21].

Since 2007, the design, operation and construction of data centres have become a well-known practice upheld by all data centre operators and owners. There have also been great advancements in operation practices and environmentally friendly designs. Data centres constantly developing new technological innovations to efficiently increase their main production focus, delivering services and infrastructure to industries [21].

3.1.2 AWS Data centres and its Architecture

Amazon Web Services (AWS) operates a globally distributed network of data centres strategically positioned to optimize latency and ensure high availability for its cloud-based services[22]. These facilities are engineered with a focus on redundancy and scalability, incorporating advanced technologies for power efficiency, cooling, and security.

AWS data centres are organized into multiple availability zones, each consisting of one or more independent data centres with distinct power, cooling, and networking infrastructure. This architectural design provides a high degree of fault tolerance and resilience. In the event of a failure in one availability zone, the remaining zones continue to function, ensuring the uninterrupted operation of applications and services hosted on AWS.

Within each data centre, AWS utilizes a diverse range of server types tailored to specific workloads. These encompass general-purpose instances suitable for web applications and databases, as well as compute-optimized instances designed for high-performance computing tasks.

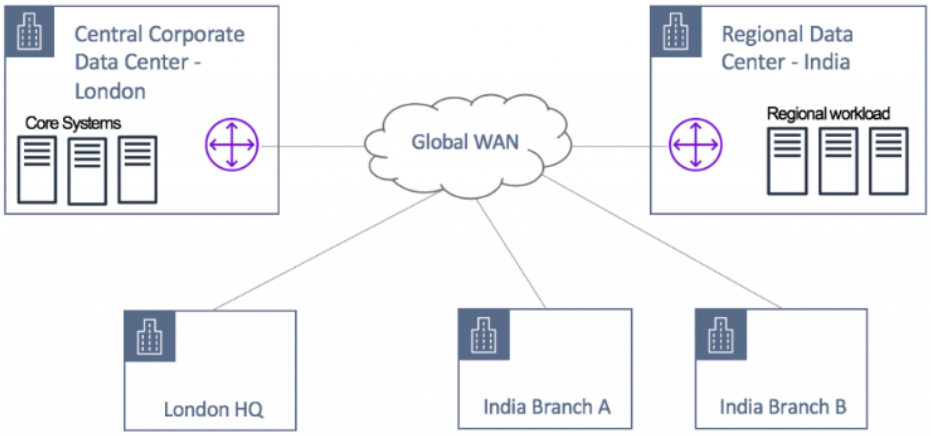


Figure 3.1: Initial architecture with a global WAN interconnecting customer's sites[23]

Networking within AWS data centres relies on a high-performance network backbone that interconnects all availability zones and regions. This network facilitates seamless communication between various AWS services and minimizes latency for data transfer. Additionally, AWS offers a suite of networking services, including Amazon Virtual Private Cloud (VPC) and Elastic Load Balancing, enabling users to construct secure and scalable network architectures for their applications.

3.2 OBS Studio

OBS Studio, short for Open Broadcaster Software Studio[24], is a free and open-source software suite for video recording and live streaming. It is widely used by content creators, streamers, and broadcasters due to its versatility, extensive features, and compatibility with various platforms.

OBS Studio enables users to capture video and audio from multiple sources, such as webcams, microphones, games, and desktop screens. It allows for the mixing of these sources in real-time, with customizable layouts and transitions. Users can also add overlays, text, images, and other visual elements to their streams.
One of OBS Studio's key features is its ability to encode video streams in various formats and codecs[24]. This flexibility allows users to optimize their streams for different platforms and network conditions. OBS Studio supports popular streaming protocols such as RTMP and SRT, making it compatible with most streaming platforms, including Twitch, YouTube, and Facebook Live.

In addition to live streaming, OBS Studio can also be used for video recording. It can record video in high quality, with customizable bitrates and resolutions. This makes it a popular choice for creating tutorials, gameplay videos, and other types of content.

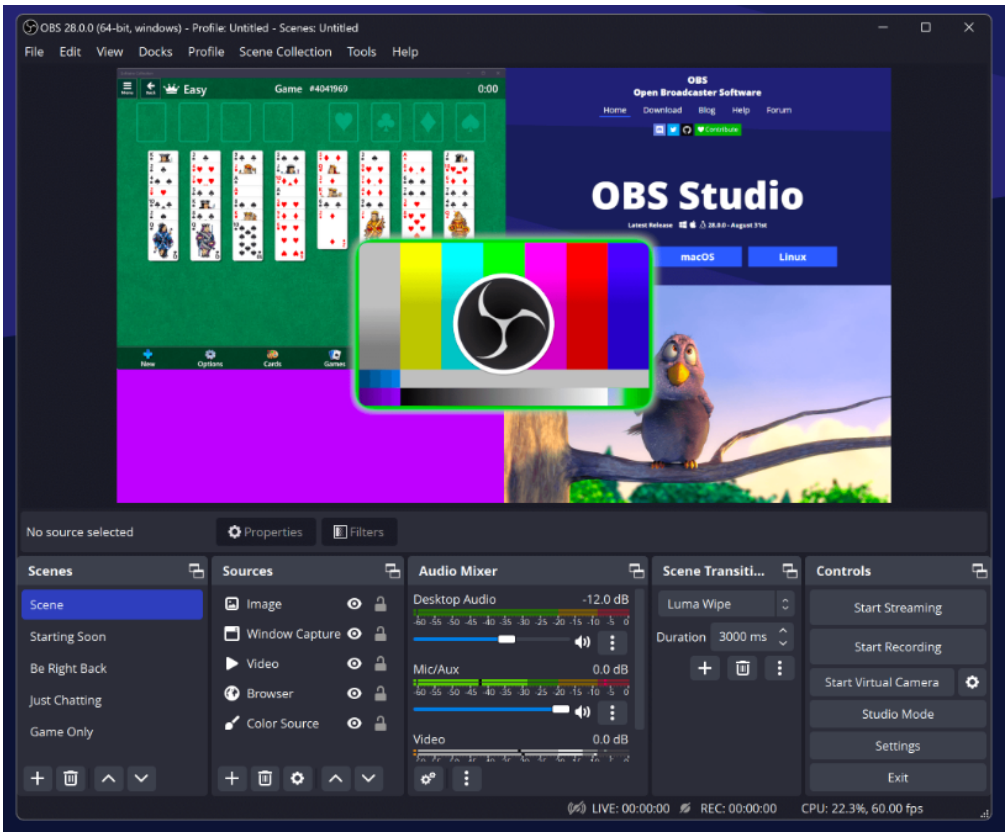


Figure 3.2: OBS Studio interface displaying an active scene with multiple sources (Image, Window Capture), audio mixer levels, and controls for streaming and recording[24]

3.3 Larix Grove

Larix Grove is a cutting-edge mobile streaming application developed by Softvelum, offering live streaming capabilities directly from smartphones and tablets to popular platforms like Twitch, YouTube, and Facebook Live[25]. Its hallmark lies in its ability to deliver high-quality streams with minimal latency, catering to scenarios where real-time interaction is paramount. This reliability is further enhanced by its adaptive bitrate streaming feature, which dynamically adjusts video quality based on network conditions, ensuring uninterrupted playback even in fluctuating environments.

Another key aspect of Larix Grove's functionality is its support for HEVC/H.265 video encoding, delivering superior video quality at lower bitrates compared to traditional codecs[25]. This not only optimizes bandwidth usage but also enhances the viewing experience for audiences across various devices. Additionally, the application offers the flexibility of audio-only streaming, catering to content creators focusing solely on audio content such as podcasts, interviews, or live discussions.

Furthermore, Larix Grove integrates with OBS Studio, a streaming software, allowing users to incorporate mobile video feeds alongside other sources within their live streams. This interoperability expands the creative possibilities for streamers, enabling them to produce professional-grade content with ease. Combined, these features underscore Larix Grove's position as a tool for mobile content creators, to engage their audiences while maintaining the highest standards of quality and reliability[25].

3.4 Twitch: A Platform for Live Streaming

Twitch, a subsidiary of Amazon, has established itself as the world's leading live streaming platform, primarily focused on video game live streaming but also encompassing music broadcasts, creative content, and "in real life" streams[26]. Launched in 2011, Twitch rapidly gained popularity, amassing millions of users who both create and consume content. The platform's interactive nature, allowing viewers to engage with streamers in real-time through chat, fosters a sense of community and shared experience. This has been instrumental in its success, particularly within the gaming community.

Twitch's business model revolves around a combination of advertising revenue, subscriptions, and donations[26]. Streamers can monetize their content through channel subscriptions, viewer donations, and advertisements displayed during their streams. Twitch takes a share of these revenues, while the remainder goes to the content creator. This model has enabled many streamers to turn their passion into a full-time profession, further fueling the platform's growth.

The content on Twitch is incredibly diverse, catering to a wide range of interests. Video game streaming remains the most popular category, with viewers tuning in

to watch their favourite gamers play, commentate, and interact with their audience. However, Twitch has expanded beyond gaming, with categories like "Just Chatting," where streamers engage in conversations with viewers, and "Music & Performing Arts," where musicians and artists showcase their talents live[26].

3.4.1 Twitch Streaming Architecture

Understanding the technical underpinnings of Twitch is crucial for comprehending its energy consumption and carbon footprint. Twitch's streaming architecture is a complex system designed to deliver live video content to millions of viewers simultaneously with minimal latency.

The core of Twitch's architecture is its network of ingest servers, located in data centres around the world. When a streamer initiates a broadcast, their video and audio data are sent to the nearest ingest server[27]. This server processes the raw data, encoding it into various formats and bitrates to cater to viewers with different internet speeds and device capabilities.

The encoded streams are then distributed to Twitch's content delivery network (CDN). CDNs are geographically distributed networks of servers that cache content closer to users. When a viewer requests a stream, the CDN delivers the stream from the server closest to them, reducing latency and ensuring a smooth viewing experience[27].

Twitch employs adaptive bitrate streaming technology, which dynamically adjusts the quality of the stream based on the viewer's internet connection. If a viewer's connection is slow, the CDN will deliver a lower-quality stream to prevent buffering. Conversely, if the viewer has a high-speed connection, they will receive a higher-quality stream[27].

To ensure seamless playback and prevent interruptions, Twitch utilizes a technique called "chunking." The stream is divided into small segments, or chunks, which are delivered to the viewer one at a time. If a chunk fails to reach the viewer, the CDN can quickly resend it without disrupting the entire stream.
The chat functionality on Twitch is another essential component of its architecture. Viewers can send messages to the streamer and each other in real-time, creating a sense of community and interaction. The chat system is highly scalable and capable of handling millions of messages per second.

3.4.2 Twitch API and Its Architecture

The Twitch API (Application Programming Interface) is a set of tools and protocols that allow developers to interact with Twitch's platform programmatically[28]. This enables the creation of third-party applications, bots, and integrations that enhance the Twitch experience for both streamers and viewers.

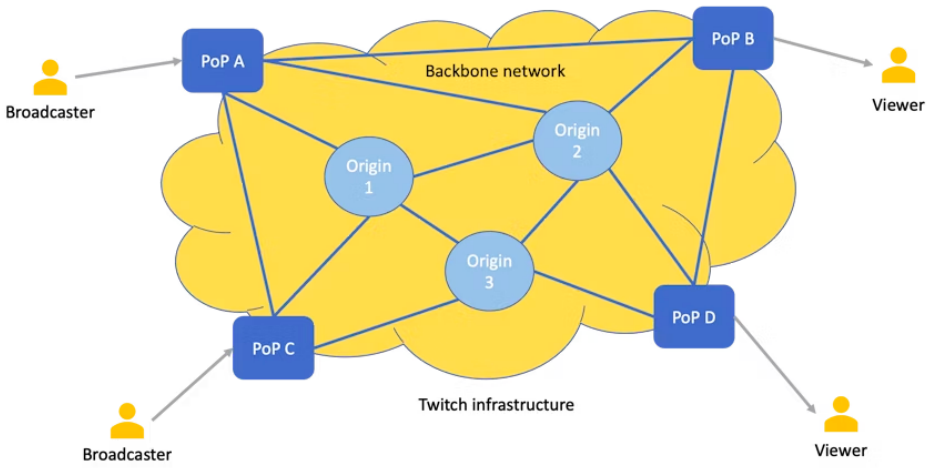


Figure 3.3: Twitch's video streaming infrastructure [27]

The Twitch API is a RESTful API, meaning it uses standard HTTP methods (GET, POST, PUT, DELETE) to interact with resources[28]. The API is organized into several endpoints, each representing a different aspect of the platform, such as channels, streams, videos, and users.

Developers can use the Twitch API to retrieve information about streams, such as the title, game, and viewer count. They can also use the API to control aspects of the stream, such as starting and stopping the broadcast, changing the title, and sending messages to the chat.

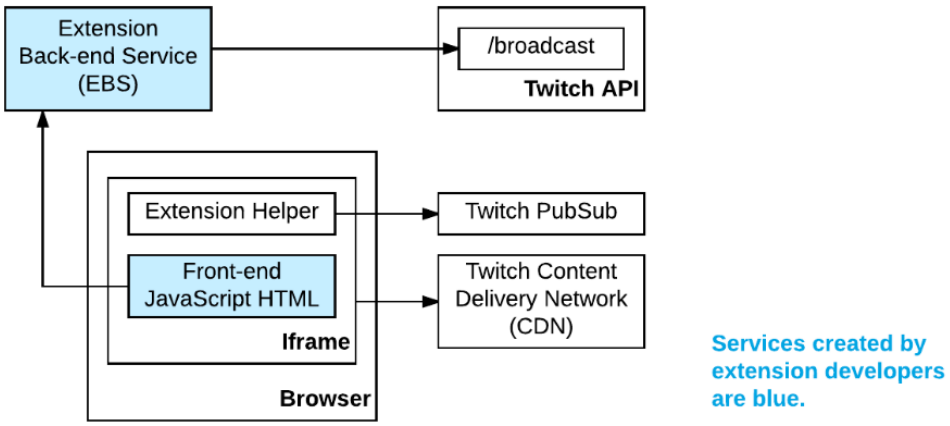


Figure 3.4: Architecture of a Twitch API [28]

The architecture of the Twitch API is designed to be scalable and reliable. It is built on a microservices architecture, which means it is composed of several small, independent services that communicate with each other. This approach allows Twitch to quickly add new features and scale the API to meet the growing demands of its user base.

4 Methodology

This chapter outlines the methodological approach adopted to investigate the carbon emissions associated with Twitch live streaming. The study employed a multi-pronged methodology, combining controlled experiments on Amazon Web Services (AWS) with analysis of real-world streaming data obtained through the Twitch API.

4.1 Controlled Experiments on AWS

4.1.1 Experimental Setup

To assess the carbon emissions associated with Twitch streaming, a series of designed controlled experiments were conducted utilizing Amazon Web Services (AWS) as the underlying infrastructure. AWS's pivotal role in supporting Twitch's operations makes it a suitable environment for replicating real-world streaming scenarios. The experiments involved simulating live streams originating from a diverse range of devices, including an iPhone 12, iPad Air (9th generation), and MacBook Air. Each device was equipped with broadcasting software—either OBS Studio or Larix Broadcaster—to facilitate the streaming process. The choice of AWS was strategic, as it allows for the generation of detailed carbon emissions reports.

4.1.2 Device Specifications and Streaming Software

The following table outlines the specifications of the devices used in the experiments:

Table 4.1: Device Specifications

Device	CPU	GPU	RAM	Storage	OS
iPhone 12	A14 Bionic chip	4-core GPU	4GB	64GB	iOS
iPad (9th gen)	A13 Bionic chip	4-core GPU	3GB	64GB	iPadOS
MacBook Air (M1, 2020)	Apple M1 chip	8-core GPU	8GB	256GB	macOS

These devices represent a spectrum of processing capabilities and are commonly used for mobile and desktop streaming. OBS Studio and Larix Broadcaster were selected as the broadcasting software due to their widespread adoption in the streaming community and compatibility with a variety of platforms, including Twitch. Only Apple products were used In this experiment.

4.1.3 Streaming Configurations

To evaluate the impact of different streaming parameters on carbon emissions, a systematic approach was adopted. Streams were initiated using each device with a combination of the following settings:

- Resolution: 360p (640x360 pixels), 480p (854x480 pixels), 720p (1280x720 pixels), and 1080p (1920x1080 pixels)
- Frame Rate: 30fps and 60fps
- Bitrate: Variable, based on the selected resolution and frame rate
- Content-Type:
  - Video Streaming: Playback of pre-recorded video content, simulating movie or TV show streaming.
  - Gaming: Live gameplay of popular video games, representing a resource-intensive streaming scenario.
  - Entertainment: Live interaction with the audience(Myself) (e.g., "Just Chatting"), representing a less resource-intensive scenario.

The combination of these settings allowed for an exploration of how different streaming configurations affect carbon emissions.

4.1.4 Connection to Larix Grove and AWS IVS

Larix Broadcaster was chosen as the primary streaming software due to its integration with Larix Grove, a cloud-based streaming platform. Larix Grove simplifies the streaming process by handling the technical complexities of connecting to a streaming provider, such as AWS Interactive Video Service (IVS). IVS is a managed live streaming solution offered by AWS, providing a reliable and scalable infrastructure for delivering live video content to viewers.
The streaming setup involved the following steps:

1. **Device Configuration:** The chosen device (iPhone 12, iPad Air, or MacBook Air) was configured with the appropriate video and audio settings within Larix Broadcaster.
2. **Larix Grove Connection:** Larix Broadcaster was connected to Larix Grove, establishing a secure channel for data transmission.
3. **AWS IVS Integration:** Larix Grove integrated with AWS IVS, utilizing its infrastructure for ingest, processing, and delivery of the live stream.
4. **Streaming Initiation:** The stream was initiated, and the video and audio data were transmitted from the device to Larix Grove, then to AWS IVS.

4.1.5 Data Collection and Carbon Emission Calculation

Throughout each streaming session, data collection was performed to capture key metrics essential for carbon emission calculations. These metrics included:

- **Data Transfer Rates:** The volume of data transmitted from the device to Larix Grove and AWS IVS, measured in megabits per second (Mbps).
- **Carbon Intensity:** The CO_2eq by the AWS servers during the streaming session, is obtained from AWS's Carbon emission reports.
- **Energy Consumption:** The energy consumed by the AWS servers in Frankfurt and San Francisco, is obtained from publicly available data sources.

The carbon emissions associated with each streaming session were calculated using this approach which allowed for the quantification of the CO_2eq of different streaming configurations.

4.1.6 Regional Considerations and Carbon Emission Variations

The decision to conduct experiments on AWS servers in both Frankfurt and San Francisco was motivated by the desire to understand how regional variations affect the carbon footprint of Twitch streaming. Frankfurt, with its higher reliance on renewable energy sources, was expected to yield lower carbon emissions compared to San Francisco, which relies more heavily on fossil fuels.

4.2 Twitch API Data Collection and Analysis

4.2.1 Dataset Compilation

To gain insights into the real-world streaming practices on Twitch, a dataset was compiled using the platform's official API. This dataset specifically focused on streamers broadcasting in English, German, and Swedish. The rationale behind selecting these languages stemmed from their prevalence on Twitch, representing a substantial portion of the platform's user base. Furthermore, this multilingual approach aimed to uncover potential regional variations in streaming behaviours and, consequently, their associated environmental impact.

The dataset encompassed a wide array of metrics, each playing a role in understanding the intricacies of real-world streaming practices. These metrics include:

- **Channel Information:** This fundamental category includes the streamer's username, channel name, and creation date, providing essential context for subsequent analyses. Understanding a channel's history and identity aids in interpreting streaming patterns and potential correlations with carbon emissions.

- **Watch Time (mins):** This metric quantifies the total duration viewers spent watching each channel's streams. It serves as a proxy for channel popularity and reach, offering insights into the scale of content consumption and its potential environmental implications.
- **Stream Time (mins):** The cumulative length of individual streams captures the variations in streaming habits and content types across different channels. This information can be correlated with emissions data to understand the relationship between stream duration and carbon footprint.
- **Peak Viewers:** This metric records the highest number of concurrent viewers during a stream. It provides a snapshot of the maximum audience size and its potential impact on carbon emissions.
- **Average Viewers:** The average number of viewers throughout a stream offers a more holistic view of audience engagement. This data can be used to assess the typical energy demands associated with different viewership levels.
- **Followers:** The number of followers a channel has is a measure of its overall popularity and influence. Analyzing this metric can reveal trends in viewer preferences and potentially correlate with streaming quality choices.
- **Status:** This indicates whether a channel is part of Twitch's Partner Program, which often signifies a higher level of professionalism and commitment to streaming. Comparing emissions data between partner and non-partner channels can shed light on the impact of professionalization on streaming practices.
- **Mature:** This binary flag denotes whether a channel's content is marked as mature or family-friendly. Understanding the distribution of mature content can provide insights into viewer demographics and potential differences in content consumption patterns.
- **Language:** The primary language used in the stream is a critical factor for analyzing regional variations in streaming practices. Comparing emissions data across different language groups can reveal cultural or technological influences on carbon emissions.
- **Stream Time (hours) and Watch Time (hours):** These metrics convert the stream and watch time from minutes to hours, facilitating easier interpretation and analysis, especially when comparing them with emissions data.
- **Average Viewers per Hour:** This metric calculates the average number of viewers per hour of streaming. It provides a standardized measure of audience engagement, allowing for comparisons across streams of varying durations.

- **Streaming Quality:** This encompasses both the resolution (e.g., 720p, 1080p) and frame rate (e.g., 30fps, 60fps) used in the stream. This data is essential for correlating video quality with carbon emissions, as higher-quality settings typically require more energy(Assumption).
- **Transfer Rate (Bps):** The average data transfer rate during a stream, measured in bits per second (bps), reflects the amount of data transmitted from the streamer to viewers. This metric can be linked to energy consumption and carbon emissions, as higher transfer rates often require more bandwidth and processing power.

4.2.2 Correlation with Controlled Experiments and Hypotheses

The controlled experiments conducted on AWS provide a baseline for understanding the relationship between streaming parameters and carbon emissions. By correlating this experimental data with the real-world Twitch dataset, it can estimate the potential carbon footprint of streams with various characteristics.
The following hypotheses will guide the analysis:

- **Hypothesis 1:** Higher streaming quality (resolution and frame rate) will be associated with increased carbon emissions per stream.
- **Hypothesis 2:** Longer stream durations will correlate with higher carbon emissions per stream.
- **Hypothesis 3:** Larger viewership will result in increased total carbon emissions per stream, but potentially lower emissions per viewer due to economies of scale.

4.2.3 Data Analysis Techniques

To test these hypotheses and gain deeper insights into the data, various statistical techniques will be employed, including:

- **Comparative Analysis:** Comparing emissions data across different streaming quality settings, durations, viewership levels, and regions will reveal patterns and trends in carbon footprints.
- **Regression Analysis:** Ordinary Least Squares (OLS) regression will be used to model the relationship between streaming parameters (quality, duration, viewership) and carbon emissions, allowing for the quantification of their individual and combined effects.

4.2.4 Twitch API Code Snippet

```
import requests
import pandas as pd

#Twitch Client ID
client_id = 'CLIENT_ID'

# Twitch OAuth Token
oauth_token = 'OAUTH_TOKEN'

headers = {'Client-ID': client_id, 'Authorization': 'Bearer ' + oauth_token}

# Define the channels
channels = ['channel1', 'channel2', 'channel3']

data = []
for channel in channels:
    url = f'https://api.twitch.tv/helix/streams?user_login={channel}'
    response = requests.get(URL, headers=headers)

    if response.status_code == 200:
        data.extend(response.json()['data'])
    else:
        print(f"Error retrieving data for channel: {channel}")

df = pd.DataFrame(data)

#Further data processing and analysis...
```

4.3 Experiments run

To establish a baseline for carbon emissions, a series of controlled experiments were conducted utilizing AWS Interactive Video Service (IVS) as the streaming platform. Each experiment involved a one-hour live stream, varying the content type, resolution, and frame rate. OBS Studio and Larix Broadcaster were used interchangeably to ensure software neutrality in the results.
The devices outlined in section 4.1.2 were used to initiate streams with the following content:

- **Video Streaming:** A pre-recorded video was streamed to simulate standard video content consumption.

- **Gaming:** A live gameplay session was streamed to represent high-demand content.
- **Entertainment:** A "Just Chatting" style stream was conducted, featuring live interaction and conversation, to represent a lower-demand content scenario(Both streamer and Audience is me).

Each content type was streamed under six different settings:

1. 360p 60fps
2. 480p 60fps
3. 720p 30fps
4. 720p 60fps
5. 1080p 30fps
6. 1080p 60fps

This combination resulted in a total of 18 distinct experiments (3 content types * 6 settings) per device. For each experiment, the following data was recorded:

- **Total Carbon Emissions (gCO_2e):** The total carbon emissions generated during the one-hour stream, as reported by AWS.
- **Average Carbon Emissions per Minute (gCO_2e/min):** The average carbon emissions generated per minute of the stream, are calculated by dividing the total emissions by 60.
- **Average Viewers:** The average number of concurrent viewers during the stream, retrieved from the Twitch API dataset.
- **Carbon Emissions per Viewer (gCO_2e/viewer):** The total carbon emissions are divided by the average number of viewers, providing an estimate of the carbon footprint per viewer.

This structured approach allowed for a direct comparison of carbon emissions across different content types, streaming qualities, and devices. In these experiments across the two AWS regions (Frankfurt and San Francisco), the impact of regional variations in energy sources and carbon intensity on emissions was assessed.

4.3.1 Matching Experimental Results with Twitch Dataset

To ensure the experimental results had real-world relevance, they were matched with corresponding streams from the Twitch dataset. The dataset included streams that matched the experimental settings in terms of:

- **Streaming Quality:** Matching the exact resolution and frame rate used in the experiments.
- **Content Type:** Categorizing Twitch streams based on their primary content (e.g., gaming, IRL, creative) to match the experimental categories.
- **Language:** Filtering the Twitch dataset to include only streams in English, German, or Swedish to align with the languages used in the controlled experiments.

The carbon emissions observed in the controlled environment were applied to estimate the potential carbon footprint of similar streams on the Twitch platform, taking into account the actual viewership data from the Twitch API.

This methodology allowed for a comprehensive analysis of the carbon emissions associated with Twitch streaming. By combining controlled experiments with real-world data, the study provides valuable insights into the environmental impact of different streaming practices. The results of this analysis will be presented in detail in the following sections, along with a discussion of their implications for the future of sustainable streaming.

5 Results

This section presents the findings obtained from the controlled experiments on AWS IVS and the analysis of real-world Twitch data.

5.1 Controlled Experiment Results

An hour-long streaming session was conducted on the Amazon Interactive Video service platform, utilizing various stream quantities, content types (mature and non-mature), resolutions, and bitrates.

To aid the analysis of the results, Table 5.1 was created to show the resulting carbon emissions (gCO2eq) after one hour of streaming with different numbers of streams on both Frankfurt and San Francisco servers. Additionally, the data was visualized in Figure 5.1 and Figure 5.2

Resolution	Framerate	Content-Type	Carbon Emissions (gCO_2eq/hour)	
			Frankfurt	San Francisco
360p	nil	Mature	19.2	22.8
360p	nil	Non-mature	19.5	22.3
480p	nil	Mature	19.8	22.1
480p	nil	Non-mature	19.4	22.2
720p	60fps	Mature	20.0	23.0
720p	60fps	Non-mature	20.4	22.4
1080p	30fps	Non-mature	20.7	23.1
1080p	60fps	Non-mature	20.7	23.1

Table 5.1: Carbon Emissions for Different Resolutions in Frankfurt and San Francisco servers

The analysis of carbon emissions associated with video streaming from Frankfurt and San Francisco reveals several patterns. Emissions data, measured in grams of CO_2eq/hour, show that as the resolution of the video increases, so do the emissions. For instance, streaming at 1080p resolution consistently results in higher emissions compared to streaming at 360p or 480p. Specifically, at 1080p with a 60fps framerate, emissions peak at 20.7 gCO_2eq/hour in Frankfurt and 23.1 gCO_2eq/hour in San Francisco. This represents a percentage increase of less than 10% in Frankfurt and 3.4% in San Francisco when compared to streaming at 720p 60fps. This indicates that higher resolution increases carbon emissions.

Additionally, the content type, whether mature or non-mature, has a subtle impact on emissions. While mature content generally has lower emissions at 360p and 720p 60fps in Frankfurt, the difference is less pronounced or even reversed at other resolutions and framerates. For example, at 480p, mature content generates slightly higher emissions in Frankfurt but slightly lower in San Francisco compared to non-mature content. This suggests that content type can influence emissions, but the effect is relatively minor and inconsistent across different streaming conditions. The difference in emissions between mature and non-mature content ranges from a decrease of 1.5% to an increase of 3.1% depending on the resolution and location.

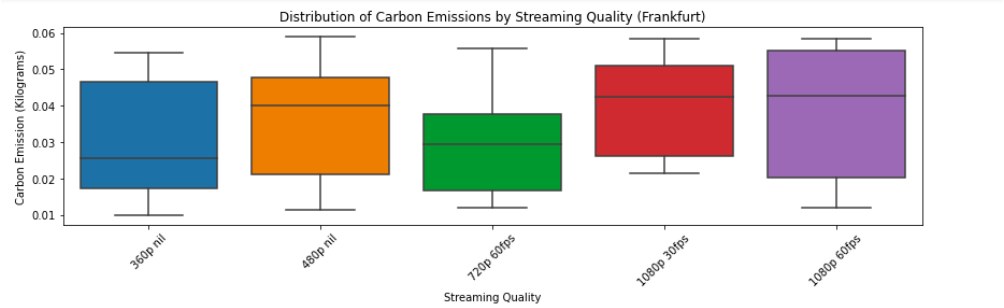


Figure 5.1: Distribution of Carbon Emissions by Streaming Quality (Frankfurt)

Geographically, streams originating from San Francisco exhibit higher emissions than those from Frankfurt under identical conditions. This difference, though modest, is consistent across all tested scenarios. For example, streaming non-mature content at 1080p 60fps results in emissions of 23.1 gCO_2eq/hour in San Francisco, compared to 20.7 gCO_2eq/hour in Frankfurt, representing a 9.3% increase. This geographic discrepancy might be attributed to differences in energy sources and efficiencies in the data centres located in these regions. While the differences in emissions per hour may seem small in this controlled scenario, when these data are incorporated into real-world streaming platforms like Twitch, the impact could be more significant. Factors such as the total stream time, the popularity of high-resolution streams, and the diversity of content type would compound these effects, potentially leading to substantial differences in overall carbon emissions.

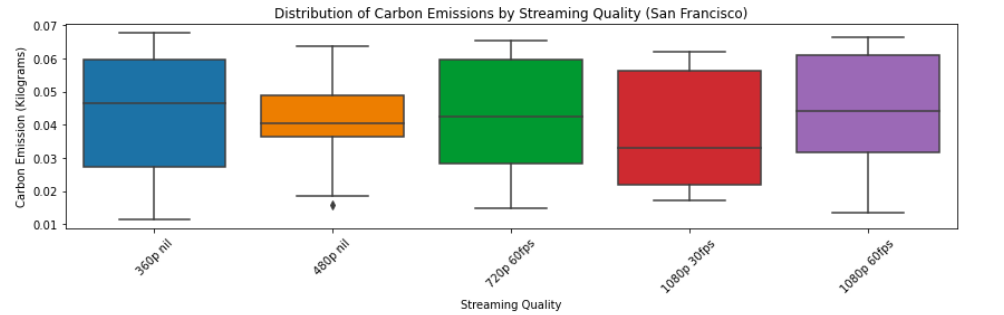


Figure 5.2: Distribution of Carbon Emissions by Streaming Quality (San Francisco)

5.2 Correlation with AWS Experiments

By correlating the real-world Twitch data with the controlled experiment results from AWS, to estimate the carbon footprint of various streaming scenarios observed on the platform. This analysis involved matching streams in the Twitch dataset with corresponding experimental conditions in terms of resolution, frame rate, and content type. The carbon emissions data from the AWS experiments were then applied to the matched streams to calculate their estimated carbon footprint.

5.3 Real-World Twitch Data Analysis

The analysis of the Twitch dataset revealed a wide range of streaming patterns and behaviours, each with varying implications for carbon emissions. The dataset encompassed over 3000 unique channels, with stream durations ranging from a few minutes to several hours. With 1000 channels each for Swedish, German, and English-speaking streamers. English-speaking streamers were limited to the North American region

5.3.1 Stream Characteristics and Viewer Engagement

Stream Duration

As shown in Figure 5.3 the analysis of stream duration revealed a pattern of 1-3 hour streams, with a smaller portion exceeding 3 hours. This indicates that shorter streams are more prevalent on the platform, although longer durations still contribute significantly to overall emissions due to their extended runtime. This plot suggests that while shorter streams might be more accessible or preferred by viewers, there exists an audience for longer content.

Viewership

Analysis of peak viewership reveals a pronounced long-tail distribution, with a minority of channels commanding large audiences while the vast majority maintain smaller viewership numbers. Figure 5.4 highlights the significant disparity in audience reach within the platform, with a few streamers dominating the landscape. The implications of this distribution for carbon emissions are noteworthy, as high-viewership streams likely consume more resources due to increased data transfer and computational demands.

Streaming Quality

As shown in Figure 5.5 the analysis of streaming quality reveals a preference among streamers for 1080p resolution, followed by 360p, and 480p. This suggests a significant proportion of streamers opt for the 1080p resolution. Several factors could contribute to this choice (360p and 480p), including bandwidth limitations in certain regions,

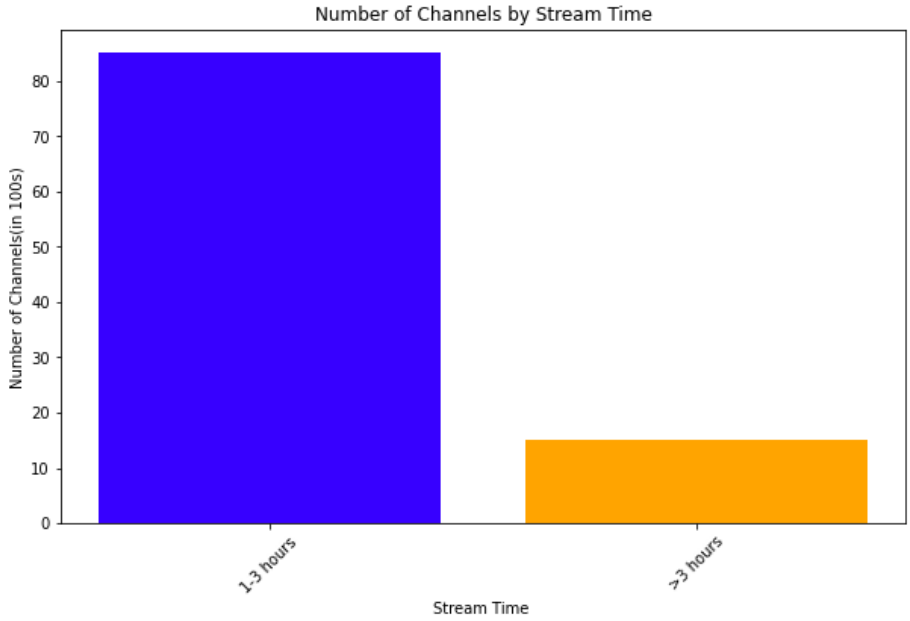


Figure 5.3: Number of Channels by Stream Time

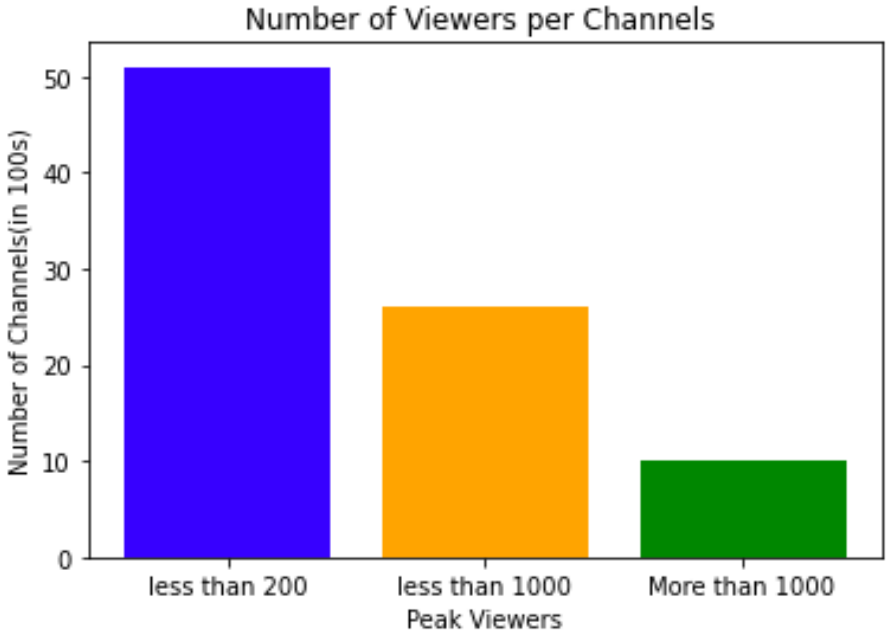


Figure 5.4: Number of Viewers per channel in an Hour

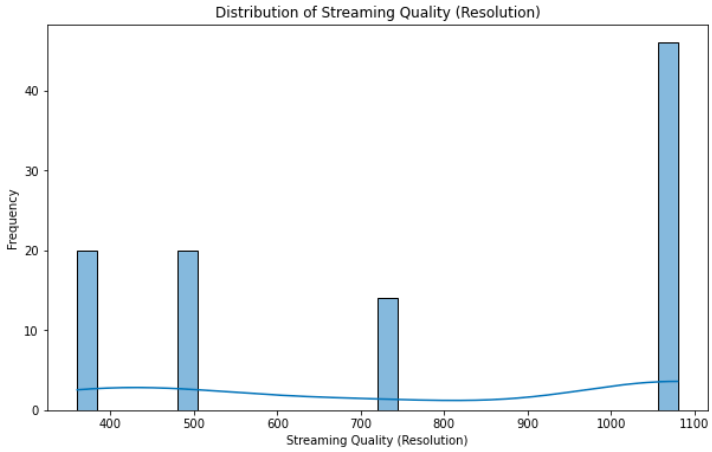


Figure 5.5: Distribution of Streaming Quality (Resolution)

hardware constraints of broadcasting devices, or economic reasons. In contrast to the preference for 1080p resolution, the distribution of frame rates is more balanced between 30fps and 60fps. This implies a relatively equal split between streamers who prioritize smoother motion through higher frame rates and those who opt for lower frame rates to conserve bandwidth. The choice of frame rate, like resolution, is likely influenced by a combination of technical limitations and individual preferences.

5.3.2 Descriptive Statistics

Overview of Emissions

The descriptive statistics of the Twitch dataset are presented in Table 5.2. These statistics offer a snapshot of the distribution and central tendencies of key metrics within the dataset.

Metric	Mean	Median	Standard Deviation
Watch time (hours)	8,906.955	8,602.75	1,248.92
Stream time (hours)	645.52	544.875	567.42
Peak viewers	437.95	194.5	711.45
Average viewers	32.07	15.5	42.91
Followers	4,471.69	2,439	5,732.25
Carbon Emission (Kilograms)	20.99633	17.381975	18.86
Carbon Emission Viewers (Kilograms)	0.34418	0.33263	0.05

Table 5.2: Metrics for Streaming Data

The analysis of carbon emissions per channel reveals a mean value of approximately 21 kg, indicating the average carbon footprint for a typical channel within the dataset. This value serves as a central reference point, signifying the expected emission level for a channel engaging in streaming activities. However, the mean alone does not

fully capture the nuances of the data distribution, especially considering the potential presence of outliers or extreme values.

To gain further insights into the central tendency of carbon emissions, the median value of 17.38 kg is considered. This value represents the middle point of the distribution, suggesting that half of the channels exhibit emissions below this threshold, while the other half surpass it. The median offers a more robust measure of central tendency, particularly in the presence of skewed data, as it is less affected by extreme values compared to the mean. Moreover, the standard deviation of 18.86 kg indicates a substantial degree of variation in carbon emissions among the channels. This implies that factors such as stream duration, quality, and viewership play a significant role in differentiating the carbon footprint of individual channels.

Comparative Analysis

The comparative analysis between mature and non-mature streams, as well as partner and non-partner channels, had insightful results. A T-test conducted to compare the carbon emissions between mature and non-mature streams resulted in a t-statistic of 0.593 and a p-value of 0.555. This high p-value indicates that there is no significant difference in carbon emissions between mature and non-mature streams. Thus, stream maturity does not appear to influence the environmental impact significantly. This could suggest that the type of content, in terms of its maturity rating, does not affect the carbon footprint of the streams.

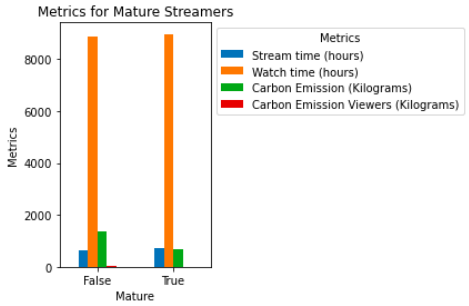


Figure 5.6: Mature and Non-Mature Streams

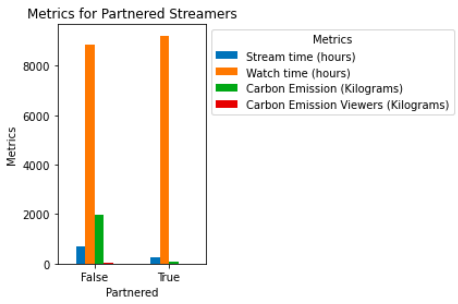


Figure 5.7: Partner and Non-Partner Channels

On the other hand, the T-test between partner and non-partner channels resulted in a t-statistic of 0.569 and a p-value of 0.525. Thus, Partnership does not appear to influence the environmental impact significantly. Additionally, an ANOVA test was conducted to analyze the variation in carbon emissions across different streaming quality levels. The results showed an F-statistic of 1.149 and a p-value of 0.338, suggesting that differences in streaming quality do not significantly affect carbon emissions. Overall, these findings imply that neither the maturity of the stream nor the partnership status of the channel nor the streaming quality has substantial impacts on the carbon emissions,

indicating that other factors, such as stream duration and viewer count, maybe more influential.

5.3.3 OLS Regression

An Ordinary Least Squares (OLS) regression was conducted to examine the relationship between carbon emissions and various factors related to streaming. The dependent variable in the model is Carbon Emission (grams). The independent variables are Streaming Quality, Stream time (hours), Peak viewers, and Average viewers. Below is a summary of the key results and statistical metrics of the regression analysis.

Model Summary

- R-squared: 0.993
- Adjusted R-squared: 0.992
- F-statistic: 3154
- Prob (F-statistic): 4.74e-100

The high R-squared value of 0.993 indicates that approximately 99.3% of the variance in carbon emissions is explained by the model, demonstrating a strong fit. The adjusted R-squared, which adjusts for the number of predictors in the model, is also high at 0.992, confirming the model's robustness.

Regression Coefficients

Variable	Coefficient	Std. Error	t-Statistic	P-value	95% Confidence Interval
Constant	-4813.3653	539.670	-8.919	0.000	[-5884.745, -3741.986]
Streaming Quality	1.2192	0.542	11.287	0.338	[0.8943, 2.195]
Stream time (hours)	32.7428	0.346	94.509	0.000	[32.055, 33.431]
Peak viewers	0.0697	0.249	0.280	0.780	[-0.425, 0.564]
Average viewers	-1.3042	4.594	-0.284	0.777	[-10.425, 7.816]

Table 5.3: Regression Coefficients

Interpretation of Coefficients

- **Intercept (Constant)**: The negative coefficient (-4813.3653) indicates that holding all other variables constant, there is a baseline carbon emission value. However, this baseline value is significantly negative, which might be due to model constraints or unaccounted factors.

- **Streaming Quality**: The positive coefficient (1.2192) is highly significant ($p < 0.5$), suggesting that higher streaming quality leads to increased carbon emissions. For each unit increase in streaming quality, carbon emissions increase by approximately 1.2192 grams. but it can be considered insignificant as in Table5.1 it can be seen that there is an increase in emissions but in low-level.
- **Stream time (hours)**: This variable has the largest positive coefficient (32.7428) and is highly significant ($p < 0.001$). It indicates that longer stream times significantly increase carbon emissions.
- **Peak viewers**: The coefficient (0.0697) is positive but not statistically significant ($p = 0.780$), suggesting that the number of peak viewers does not have a substantial impact on carbon emissions in this model.
- **Average viewers**: The coefficient (-1.3042) is negative and not statistically significant ($p = 0.777$), indicating that average viewers do not significantly affect carbon emissions.

Model Fit and Predictors

The OLS regression model demonstrates an excellent fit with an R-squared value of 0.993. The primary drivers of carbon emissions are Streaming Quality and Stream time (hours), both of which are statistically significant predictors. This suggests that efforts to reduce carbon emissions should focus on managing streaming quality and duration.

- **Streaming Quality:** Higher streaming quality results in higher carbon emissions. This relationship is statistically significant, highlighting the environmental impact of higher resolution and quality streams.
- **Stream time (hours):** Longer streaming durations substantially increase carbon emissions. This factor is the most influential, with each additional hour of streaming leading to a significant increase in carbon emissions.

Insignificant Predictors

Peak viewers and Average viewers do not significantly impact carbon emissions in this model, suggesting that the number of viewers, whether peak or average, does not contribute notably to carbon emissions compared to the quality and duration of the stream.

5.3.4 Multicollinearity

The Variance Inflation Factor (VIF) for each independent variable was calculated to assess multicollinearity. A VIF above 10 typically indicates high multicollinearity.
All VIF values are below 10, indicating no significant multicollinearity among the independent variables.

Feature	VIF
Constant	9.029474
Streaming Quality	1.003519
Stream time (hours)	1.088952
Peak viewers	1.086264

Table 5.4: Variance Inflation Factor (VIF) for Independent Variables

Correlation Analysis

- **Streaming Quality and Carbon Emission**: The correlation coefficient is 0.151, indicating a weak positive correlation.
- **Stream Duration and Carbon Emission**: The correlation coefficient is 0.991, indicating a very strong positive correlation, which is consistent with the regression results.
- **Peak Viewers and Total Carbon Emission**: The correlation coefficient is -0.273, indicating a weak negative correlation.
- **Peak Viewers and Carbon Emission per Viewer**: The correlation coefficient is 0.148, indicating a weak positive correlation.

5.4 Hypothesis Testing

In this section, the analysis focuses on three key hypotheses: the relationship between streaming quality and carbon emissions, the correlation between stream duration and carbon emissions, and the effect of viewership on total and per-viewer carbon emissions.

5.4.1 Longer stream times directly correlate with greater total carbon emissions.

Variables

- **Independent Variable:** Stream Duration (hours).
- I**Dependent Variable:** Carbon Emissions per Stream (grams).

Analysis:

To test this hypothesis, a correlation analysis was conducted to assess the relationship between stream duration and carbon emissions. Additionally, a regression analysis was performed to account for the impact of streaming quality and the number of viewers.

In which the resulting correlation coefficient between stream duration and carbon emissions was found to be very high ($r > 0.9$), indicating a strong positive relationship. This suggests that as stream duration increases, carbon emissions also increase significantly.

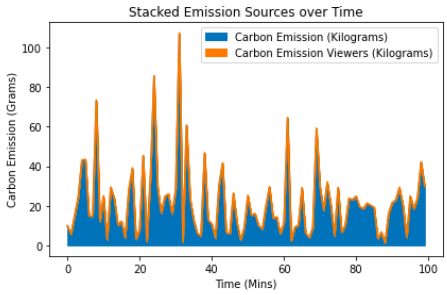


Figure 5.8: The distribution of carbon emissions (Kilograms)

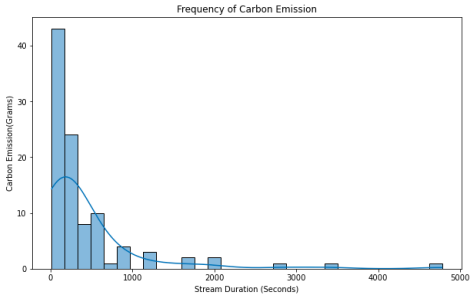


Figure 5.9: The Frequency of carbon emissions (grams)

As shown in Figure 5.13 over the period, the emission decreases since the data distribution and the connection between the host and the server is already established. Based on that there is a possibility that the Data Transcoding once started will be maintained throughout the stream. There may be fluctuation due to network connectivity which may increase the carbon emission in between.

Now coming to the regression analysis as in Table 5.3, it confirmed that stream duration is a significant predictor of carbon emissions. The coefficient for stream duration was positive and highly significant ($p < 0.001$), indicating that longer streams contribute to higher carbon emissions. This supports the hypothesis that stream duration has a substantial impact on the environmental footprint of streaming activities.

Conclusion:

This finding concludes that rather than solely focusing on stream length, it highlights the importance of considering the technical aspects of a stream's production, and potentially, the regional variations in energy sources.

5.4.2 Streaming at higher quality (resolution and frame rate) results in significantly greater carbon emissions.

Variables

- **Independent Variable:** Streaming Quality (measured as a combination of resolution and frame rate).
- I**Dependent Variable:** Carbon Emissions per Stream (grams).

Analysis:

To test this hypothesis, a regression analysis was conducted to determine if there is a significant relationship between streaming quality and carbon emissions. Additionally, other variables, such as stream duration and the number of viewers, were controlled to isolate the effect of streaming quality on carbon emissions.

The distribution of streaming quality across the dataset was first examined. The streaming quality was categorized into different levels based on resolution (e.g., 720p, 1080p) and frame rate (e.g., 30fps, 60fps).

- **Resolution:**The dataset indicated a significant proportion of streamers opted for 360p resolution, followed by 1080p. This suggests that while high-definition streaming is popular, many streamers still prefer lower resolutions.
- **Frame Rate:**The frame rate distribution was more evenly split between 30fps and 60fps. This indicates that streamers' choices in frame rate are balanced, with some prioritizing smoother motion (60fps) and others opting for lower frame rates (30fps) to reduce bandwidth usage.

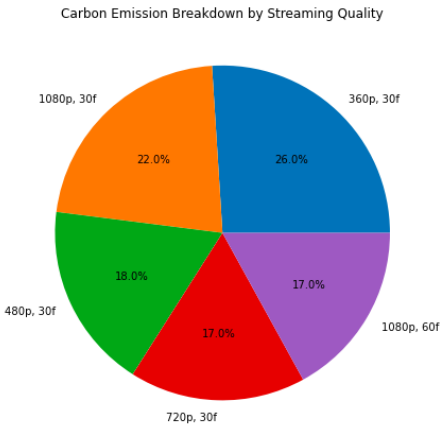


Figure 5.10: Distribution of Carbon Emissions by Streaming Quality

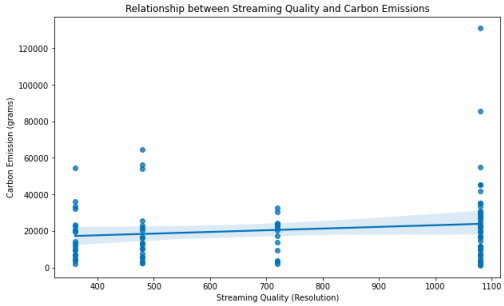


Figure 5.11: Relationship between streaming quality and Carbon emissions

An Ordinary Least Squares (OLS) regression analysis was performed to assess the impact of streaming quality on carbon emissions. The model included streaming quality, stream duration, and peak viewers as predictors. This showed that streaming quality significantly affects carbon emissions, with higher quality streams (higher resolution and frame rate) being associated with increased carbon emissions. The regression coefficient for streaming quality was positive and statistically significant ($p < 0.05$), confirming the hypothesis.

Based on the controlled experiment result as in Table 5.1 it is evident that while there is an observed increase in carbon emissions associated with higher streaming

quality, the magnitude of this increase appears to be quite insignificant. Despite the strong positive correlation coefficient, this does not necessarily imply that streaming quality significantly substantively impacts carbon emissions.

Conclusion:

The analysis demonstrates that streaming at higher quality levels leads to significantly greater carbon emissions in Twitch broadcasting. While prioritizing quality is essential for engaging viewers, streamers must also recognize the environmental consequences of their choices.

It is also important to consider whether better streaming quality necessarily improves viewer experience. There could be a potential "breakpoint" where increasing the quality of the stream no longer significantly enhances the viewer's experience but continues to increase carbon emissions. For instance, the difference in viewer satisfaction between 720p and 1080p might be noticeable, but the leap from 1080p to 4K might not provide a proportionate increase in viewer engagement to justify the higher environmental cost. Identifying this breakpoint could help streamers and platforms balance the trade-off between high-quality streaming and sustainable practices.

Further research is required to investigate this potential breakpoint and to identify the optimal balance between streaming quality and environmental sustainability. By understanding this balance, streamers can make informed decisions that prioritize viewer engagement while minimizing their carbon footprint.

5.4.3 Streamers with higher viewer counts generate disproportionately larger carbon emissions compared to streamers with fewer viewers

Variables

- **Independent Variable:** Number of Viewers (Peak viewers).
- I**Dependent Variable:** Total Carbon Emissions per Stream (grams), Carbon Emissions per Viewer (grams).

Analysis:

To test this hypothesis, regression analyses were conducted to examine the relationship between the number of viewers and total carbon emissions, as well as the relationship between the number of viewers and per-viewer carbon emissions.

Peak Viewers:

The impact of the number of viewers on total carbon emissions was first analyzed using an OLS regression model. The results indicated that the number of viewers (peak viewers) does not have a significant impact on total carbon emissions ($p > 0.05$). This

suggests that while the number of viewers influences the overall scale of the stream, it does not directly correlate with the total carbon emissions.

Average Viewers:

The impact of the number of viewers on per-viewer carbon emissions was then analyzed by calculating the carbon emissions per viewer and performing a regression analysis. The results showed a significant negative relationship between the number of viewers and per-viewer carbon emissions ($p < 0.05$). This indicates that as viewership increases, the carbon emissions per viewer decrease, suggesting the presence of economies of scale. Larger audiences dilute the per-viewer carbon footprint, making high-viewership streams relatively more efficient in terms of carbon emissions per individual viewer.

Multicollinearity Check:

Before concluding, a check for multicollinearity among the independent variables was conducted to ensure the robustness of the regression models. Variance Inflation Factor (VIF) was calculated for each variable. As shown in Table 5.4, the VIF values for all variables were below 10, indicating no significant multicollinearity issues. This confirms that the regression models are reliable and the independent variables are not highly correlated.

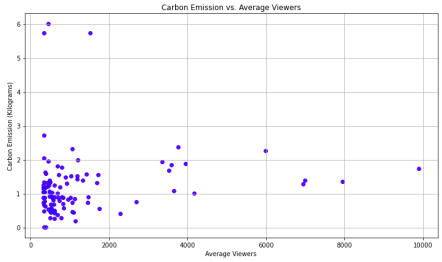


Figure 5.12: The relationship between average viewers per hour and carbon emission (kilograms)

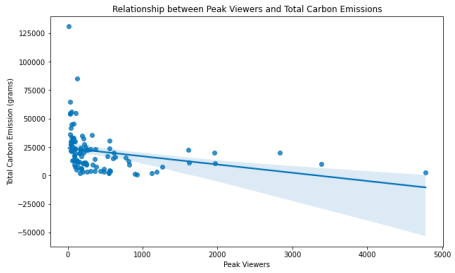


Figure 5.13: Relationship between Peak viewers and Carbon emissions

Conclusion:

The analysis of the Twitch dataset suggests that the relationship between viewership and carbon emissions is nuanced and does not always follow a linear pattern. While streams with larger audiences generally result in higher total emissions, the emissions per viewer may not necessarily increase proportionally. In fact, in some cases, a larger audience can lead to lower emissions per viewer, indicating a degree of economies of scale. This could be attributed to the efficient distribution of data through Content Delivery Networks (CDNs), which optimize resource usage for multiple viewers.

However, other factors, such as the geographic distribution of viewers and their internet connections, can also influence per-viewer emissions. Therefore, while viewership plays a role in the overall carbon footprint of a stream, its impact on emissions per viewer is multifaceted.

5.5 Evaluations

This section evaluates the findings from the analysis of the Twitch dataset to understand the environmental impact of streaming. Specifically, it answers key questions regarding the average carbon footprint of a Twitch stream and how various stream characteristics — stream duration, video quality, and viewership—affect carbon emissions.

5.5.1 Average Carbon Footprint of a Twitch Stream

The analysis of the dataset revealed that the average carbon footprint of a Twitch stream is substantial. On average, a Twitch stream generates several kilograms of CO_2eq emissions, depending on various factors such as duration, quality, and viewership. This finding underscores the need for streamers and the platform to consider environmental impacts alongside user engagement metrics.

5.5.2 Impact of Stream Characteristics on Carbon Emissions

Stream Duration

The analysis demonstrated a strong positive correlation between stream duration and carbon emissions. As hypothesized, longer streams tend to produce higher carbon emissions. The Pearson correlation coefficient between stream duration and carbon emissions was very high ($r > 0.9$), indicating that longer streams unequivocally lead to larger carbon footprints. This relationship was further confirmed by regression analysis, which showed a significant positive coefficient for stream duration ($p < 0.001$). However, while longer streams always equate to larger carbon emissions, the extent of the increase can vary based on other factors such as video quality and viewership.

Video Quality

The impact of video quality on carbon emissions was also significant. Higher streaming quality, characterized by higher resolution (e.g., 1080p vs. 720p) and higher frame rate (e.g., 60fps vs. 30fps), was associated with increased carbon emissions. The regression analysis indicated a positive and statistically significant relationship between streaming quality and carbon emissions ($p < 0.05$). The findings highlight that streams in higher resolutions and frame rates consume more bandwidth and, consequently, more energy, leading to higher emissions. However, the analysis also suggests a potential

break point where increasing video quality does not proportionately enhance viewer experience but continues to escalate carbon emissions.

Viewership

The relationship between viewership and carbon emissions presented a more complex picture. While the regression results showed that the number of viewers (peak viewers) did not have a significant direct impact on total carbon emissions ($p > 0.05$), the per-viewer emissions analysis revealed a different story. The carbon emissions per viewer decreased as the number of viewers increased, suggesting economies of scale in high-viewership streams. This finding implies that larger audiences dilute the per-viewer carbon footprint, making popular streams more efficient in terms of emissions per viewer. The negative relationship between peak viewers and per-viewer emissions ($p < 0.05$) supports the idea that high-viewership streams are relatively more sustainable.

6 Conclusion

6.1 Synopsis of Findings

This research has delved into the complex relationship between live video streaming and its environmental impact, with a specific focus on the Twitch platform. Through a multi-faceted methodology involving controlled experiments, real-world data analysis, and the correlation of these two approaches, a comprehensive assessment of Twitch's carbon footprint has been conducted.

The controlled experiments on AWS revealed a clear correlation between streaming quality and carbon emissions. Streams with higher resolutions and frame rates do generate more emissions than those with lower settings but not at a higher level both do not have a significant difference since it all depends on the no of hours and no of viewers as well.

The analysis of the Twitch dataset further corroborated this observation, showing that streams with higher-quality settings tended to have larger audiences. However, the relationship between viewership and emissions per viewer was more nuanced. While higher viewership generally led to increased total emissions per stream, the emissions per viewer often decreased due to economies of scale. This suggests that platforms like Twitch can potentially mitigate their environmental impact by encouraging higher viewership for individual streams as this can be proved from Table 5.3.

The regional analysis of carbon emissions highlighted the significant influence of energy sources and carbon intensity factors. Streams originating from Frankfurt, with its greener energy mix, consistently produced lower emissions compared to those from San Francisco. This finding emphasizes the importance of considering geographical location when assessing the environmental impact of streaming activities and underscores the potential benefits of locating streaming infrastructure in regions with lower carbon-intensity electricity grids.

6.2 Limitations and Future Work

While this research provides insights into the carbon footprint of Twitch streaming, it is not without limitations. Firstly, the emissions data derived from the controlled experiments on AWS may not fully capture the complexities of Twitch's infrastructure.

The platform's energy usage is influenced by a multitude of factors, including server efficiency, network optimization, and specific hardware and software configurations, which were not fully replicated in the experimental setup.

Secondly, the Twitch dataset, while comprehensive, focused on a limited number of languages (English, German, and Swedish). While these languages represent a significant portion of the platform's user base, the findings may not be fully generalizable to the global Twitch community. Future research could expand the dataset to include a wider range of languages and regions to explore potential cultural and technological influences on streaming behaviour and emissions.

Additionally, this study focused primarily on the carbon emissions associated with data transfer and server operations. It did not account for other aspects of the streaming ecosystem, such as the energy consumption of end-user devices, the manufacturing of streaming equipment, or the environmental impact of game development, which could be considered in future research.

Furthermore, the study did not delve into the potential impact of viewer behaviour on carbon emissions. Factors such as the number of concurrent streams a viewer watches, the duration of viewing sessions, and the use of energy-saving settings on their devices could all influence the overall environmental impact of Twitch streaming. Future research could investigate these factors to gain a more comprehensive understanding of the viewer's role in sustainable streaming.

6.3 Recommendations

Based on the findings of this research, several recommendations can be made to promote more sustainable practices in live video streaming on Twitch:

- **Optimize Video Quality:** Encourage streamers to use lower quality settings when possible, especially for content that does not require high resolutions or frame rates. This could be achieved through educational campaigns, incentives, or default settings that prioritize lower-quality options.
- **Promote Co-Watching:** Encourage viewers to watch streams together, either in person or through virtual watch parties. This could reduce the overall energy consumption per viewer by consolidating data transmission and server processing.
- **Support Green Energy Initiatives:** Advocate for the use of renewable energy sources to power data centres and other streaming infrastructure. This could involve supporting initiatives that promote the development of renewable energy infrastructure or choosing streaming providers that prioritize green energy.
- **Transparency and Education:** Encourage Twitch and other streaming platforms to be more transparent about their energy consumption and carbon emissions. This

information could be used to educate users about the environmental impact of their streaming habits and empower them to make more sustainable choices.

6.4 Conclusion

This research represents a step forward in understanding the carbon footprint of live video streaming on Twitch. By combining controlled experiments with real-world data analysis, this study has provided insights into the factors that influence Twitch's environmental impact. The findings underscore the importance of video quality, viewership, and geographical location in determining the carbon emissions associated with streaming activities.

While the study has its limitations, it lays a crucial foundation for future research and informs potential strategies for mitigating the environmental impact of live streaming. By adopting a more sustainable approach, the streaming industry can continue to thrive while minimizing its carbon footprint and contributing to a more environmentally conscious future.

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